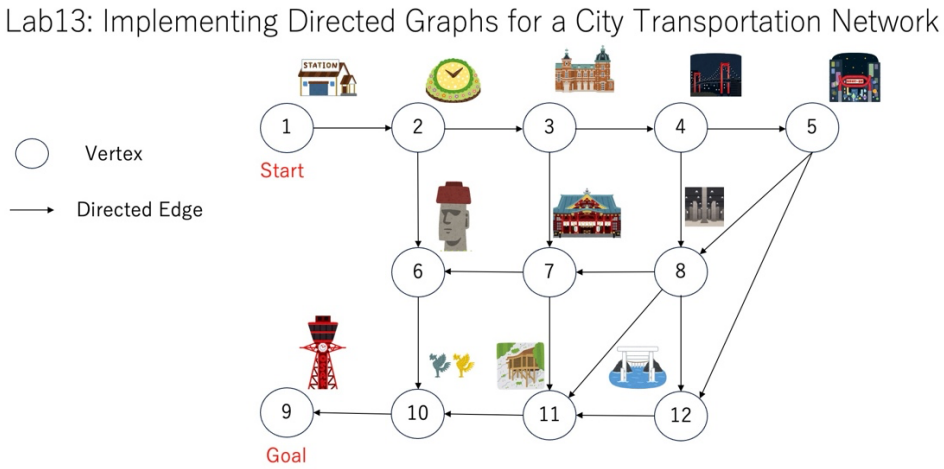
Lab Week 13, Implementing Directed Graphs for a City Transportation Network

Definition of the Graph:

The graph consists of 12 landmarks and 18 roads. This satisfies the assignment requirement of having at least 10 landmarks and 15 roads.

The traversal starts from **Vertex 1**, and the destination vertex is **Vertex 9**.

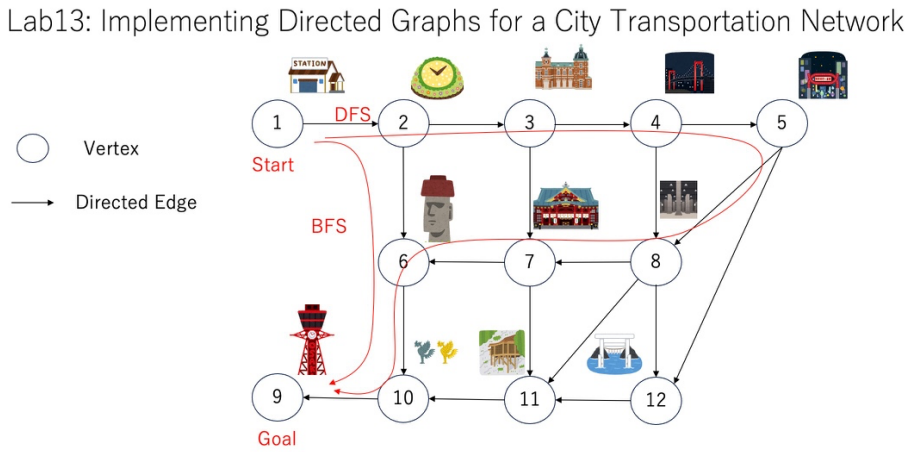


Results of BFS and DFS Traversals:

BFS result: 1 -> 2 -> 3 -> 6 -> 4 -> 7 -> 10 -> 5 -> 8 -> 11 -> 9 -> 12

DFS result: 1 -> 2 -> 3 -> 4 -> 5 -> 8 -> 7 -> 6 -> 10 -> 9 -> 11 -> 12

Shortest path using BFS: 1 -> 2 -> 6 -> 10 -> 9



BFS found the shortest path because it explores vertices level by level. In contrast, DFS does not guarantee the shortest path in a graph when multiple paths exist between vertices. As shown in this case, DFS may reach the destination through a different path that is longer than the shortest path.